

3.10 Plasma

▼ What is plasma?

- The **liquid component of the blood.**

▼ Approximately what percentage of the blood does plasma account for?

- **55%**

▼ What is serum?

- Plasma **minus the clotting factors.**

▼ How is serum obtained from plasma?

- The **plasma is collected within a tube**, and no anticoagulants are added
- This allows the **clotting factors to combine together and form a clot**, which is **removed to obtain serum**

▼ How much water is present in the average male?

- **40L.**

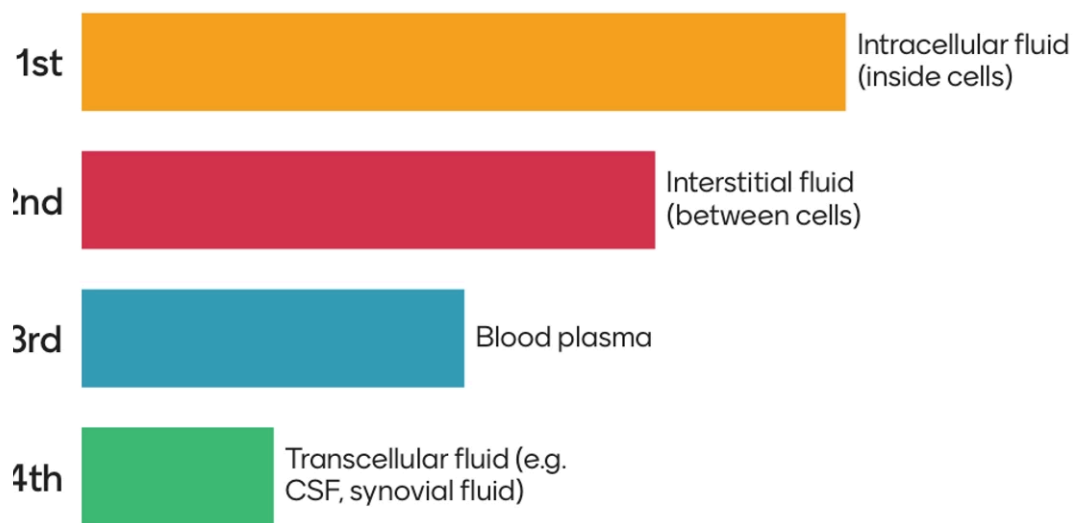
▼ What are the four main fluid components where water is found in the body?

- **Intracellular fluid**
- **Interstitial fluid (ECM fluid)**
- **Transcellular fluid (i.e CSF, synovial fluid)**
- **Plasma (blood)**

▼ Rank the prevalence of these fluid components in our bodies from highest to lowest.

- **Intracellular fluid (23L)**
- **Interstitial fluid (15L)**
- **Blood plasma (3L)**
- **Transcellular fluid (1L)**

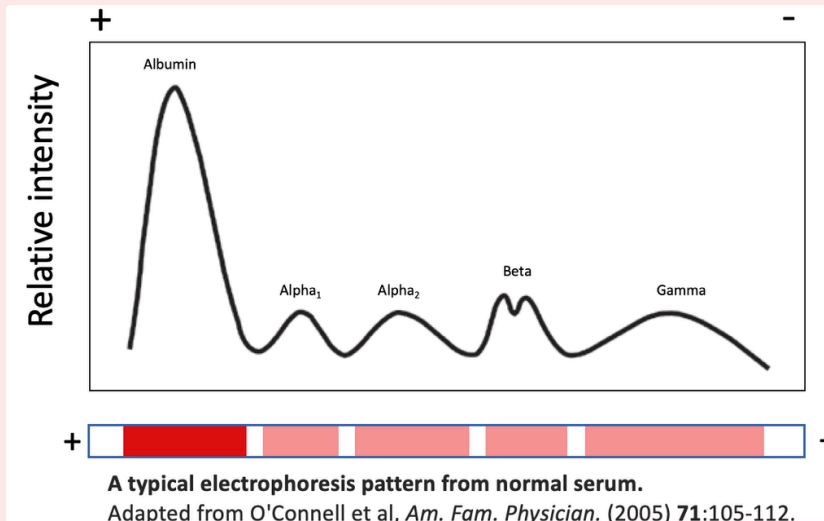
▼ Diagram showing the prevalence of the different types of fluid in our bodies.



▼ Recall five functions of plasma.

- **Clotting** - it contains **clotting factors**
- **Immunity** - it contains **antibodies and complement factors**
- **Osmotic pressure maintenance** - Proteins such as **albumin** help to **maintain colloidal osmotic pressure**
- **Metabolism** - **molecules to be metabolised** often move around in the plasma
- **Endocrine** - many **hormones** travel in plasma to the target organs
- **Excretion** - **waste products of metabolism** (i.e urea) **travel through the plasma** to the site of excretion

- ▼ What technique can be used to assess the levels of different proteins present in the plasma?
 - **Gel electrophoresis.**
 - A **patient's electrophoresis pattern** can be compared to this typical one **to assess for plasma protein deficiencies**



- ▼ What are the two major types of plasma protein?
 - **Serum albumin**
 - **Globulins**
- ▼ What is serum albumin?
 - The **most abundant (55%) protein in the plasma.**
- ▼ Recall the two roles of serum albumin.
 - **Transport of lipids, hormones and ions**
 - **Fatty acids released by lipolysis** from the breakdown of triglycerides in adipose tissue are transported by albumin for use by tissues in **β -oxidation**.
 - **Maintaining osmotic pressure of the plasma.**
 - It does this using its **colloidal osmotic pressure (COP) effect** and **drawing water into the blood**
- ▼ What is the second most abundant plasma protein type?
 - **Globulins**

▼ Recall the four main types of globulin.

- **Alpha-1 globulin**
- **Alpha-2 globulin**
- **Beta globulin**
- **Gamma globulin**

▼ Recall one alpha-1 globulin.

- **Alpha-1 antitrypsin. (A1AT)**

▼ Where is alpha-1 antitrypsin (A1AT) produced?

- **Alpha-1 antitrypsin (A1AT)** is an alpha-1 globulin **produced in the liver**

▼ What does alpha-1 antitrypsin (A1AT) do?

- **Alpha-1 antitrypsin (A1AT)** is used to **inhibit the function of proteolytic enzymes such as neutrophil elastase**, preventing excess damage to tissues

▼ What is neutrophil elastase, and what does it do?

- Neutrophil elastase is a **protease** used to **break down a key extracellular protein called elastin** which **gives tissues their elastic qualities**

▼ When is neutrophil elastase released?

- It is **released by neutrophils during inflammation**

▼ What can alpha-1 antitrypsin (A1AT) deficiency hence lead to?

- Deficient A1AT can **compromise the lung** where **degradation of lung tissue** leads to a **loss of elasticity and respiratory problems**
 - i.e COPD

▼ Describe the role of alpha-1 globulins.

- To inhibit enzymes that are involved in the breakdown of proteins (proteases)
 - i.e **neutrophil elastase** released by neutrophils during inflammation which can break down **the elastin protein**.

▼ Recall two types of alpha-2 globulin.

- **Haptoglobin**
- **Alpha-2 macroglobulin**

▼ What is the role of haptoglobin?

- It **binds onto haemoglobin released by erythrocytes** and the resulting **haptoglobin:haemoglobin complex is removed by the spleen**
- **High haptoglobin** can indicate **high haemoglobin levels** in the plasma

▼ How can haptoglobin be used clinically?

- **High haptoglobin** can indicate **high haemoglobin levels in the plasma** and hence **haemolytic anaemia**

▼ What is the role of alpha-2 macroglobulin?

- It is a **protease inhibitor** which can **stop fibrinolysis**, so that **fibrin is not broken down**
 - It does this by **inhibiting plasmin** - the principle fibrinolytic enzyme

▼ Recall two types of beta-globulin.

- **Complement C3/C4**
- **Transferrin**

▼ What is the role of transferrin?

- To **transport iron that is released from the stores of ferritin or dietary iron to wherever it is needed metabolically**

▼ Recall two types of gamma-globulin.

- **Immunoglobulins**
- **C-reactive proteins.**

▼ How can C-reactive proteins be used diagnostically?

- **Increases** in the relative amounts of **C-reactive proteins can indicate inflammation (and hence infection)**

▼ Hence, what can increases in the relative amounts of gamma-globulin indicate?

- **Gamma-globulins** include **C-reactive proteins** and **immunoglobulins**
- **Increases** in the **relative amounts of gamma-globulin** can indicate **infection, inflammation as well as myeloma** (cancer of B plasma cells)

▼ What is the most abundant cation in the plasma?

- **Na⁺**

▼ What is the most abundant cation in red blood cells?

- **K⁺**

- ▼ What mechanism in the body takes up over 1/3 of ATP at resting state?
 - The Na^+/K^+ pump.
- ▼ How much Na^+ is pumped per K^+ and in which directions?
 - 3 Na^+ pumped out for every 2 K^+ pumped in.
- ▼ Why is it important for cells that this pump keeps on operating?
 - To **maintain osmotic pressure within cells.**
 - **Lack of operation** can lead to **influx of sodium ions and hence water**, leading to **cell lysis by bursting**
 - **Cells can even shrink if extracellular Na^+ gets too high**, which also causes problems such as death
- ▼ Which ion is used as an important cofactor for many enzymes?
 - **Mg^{2+} (magnesium)**
 - When **intracellular**
- ▼ What is meant when plasma is referred to as a "biomarker"?
 - It describes how the **proteome of the plasma (levels of proteins within it)** can be analysed and how different proteins can **give us indications of certain diseases** etc.
 - i.e gamma-globulins indicate infection, myeloma etc.
 - Links with **disease, lifestyle and medications** can also be made to the proteome
- ▼ Recall the main clinical use of plasma.
 - **Passive immunotherapy**
- ▼ What is passive immunotherapy involving plasma?
 - It can be used to **transfuse plasma (or immunoglobulins directly) to patients who require higher levels of immunoglobulins**, for example
 - Antibodies can be administered intravenously to give protection against common infections.
 - This is a form of **passive immunity**, as the **antibodies are being transferred** to the recipient, in contrary to a **vaccine** which is **active immunity** as the **antibody is produced by the patient's immune system.**

- ▼ Recall the two main types of passive immunotherapy involving plasma.
 - **Intravenous Immunoglobulin G (IVIG)**
 - This protects against most common pathogens. IgG is given to patients.
 - **Hyperimmune globulin**
 - i.e a patient has too much of an immunoglobulin, so some is taken out and frozen to be given to another patient who needs it
- ▼ How can donors be selected for immunotherapy?
 - Donors can be screened for high levels of IgG against particular pathogens
 - Those with high levels can be selected as donors.
- ▼ Diagram illustrating the use of convalescent plasma (plasma from recovered patients) for COVID-19 patients